

BAYESIAN ANALYSIS

Homework II 2024

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SECTION a): How would you carry out an analysis to compare the number of points scored in a NBA's match with the number of points scored in an ACB's match? Specify which statistical model you would use and which would be the prior in both ACB and NBA games.

SOLUTION:

When modeling the number of points scored in a basketball match the possible values taken for the statistical model are integer values. The Poisson distribution seems to be a good candidate in our problem. Other discrete probability distributions, as the Negative Binomial distribution, can be used. In fact, the Poisson distribution is a limiting case of the Negative Binomial distribution. One can see the Negative Binomial distribution as an extension of the Poisson, allowing overdispersion (a phenomena occurring when the variance is greater than the mean). Anyway, in this problem we will be using the Poisson distribution. Thus, we have two statistical models, one for the NBA points which is

$$Y_1 \sim Poiss(\lambda_{NBA})$$

and another for the ACB which is

$$Y_2 \sim Poiss(\lambda_{ACB}).$$

The following step is to choose the prior distribution in both NBA and ACB games. Since it is the conjugate one, the Gamma distribution will be the chosen one for our Bayesian models. Thus, we have the priors

$$\pi(\lambda_{NBA}) \sim Gamma(\alpha_{NBA}, \beta_{NBA})$$

and

$$\pi(\lambda_{ACB}) \sim Gamma(\alpha_{ACB}, \beta_{ACB}).$$

So now, we have to choose the values $\alpha_{NBA}, \beta_{NBA}, \alpha_{ACB}, \beta_{ACB}$, depending on our prior information. Honestly, I am a loyal NBA fan. I follow the league on a daily basis and I am very informed on what is going on with the league. Nonetheless, I do not follow the ACB so much, but I also have a good prior knowledge about the Spanish league. To my belief, in the NBA more points are scored than in the ACB. However, with the aim of provide a powerful prior information to our Bayesian model, I have done some research on the web. *Basketball Reference* ^[1] is the main website for the NBA statistics. Moreover, I have consulted the ACB website ^[2] for the Spanish basketball league statistics. I found the average points per game over the last twenty seasons in both leagues. This will be our prior information on the problem. I have chosen the last twenty years because I do not know how old is the data provided in this assignment, and then take into account the scoring in the last two decades is a good source of information.

Season	NBA	ACB
2023-24	114.5	83.5
2022-23	114.7	81.2
2021-22	110.6	78.4
2020-21	112.1	77.9
2019-20	111.8	76.5
2018-19	111.2	78.1
2017-18	106.3	81.8
2016-17	105.6	79.2
2015-16	102.7	77.8
2014-15	100.0	76.9
2013-14	101.0	76.5
2012-13	98.1	77.4
2011-12	96.3	78.2
2010-11	99.6	77.9
2009-10	100.4	78.7
2008-09	100	77.8
2007-08	99.9	77.4
2006-07	98.7	78.1
2005-06	97.0	77.9
2004-05	97.2	77.3
2003-04	93.4	78.3

Table 1: Average points per game over the last twenty seasons.

¹See the website on <https://www.basketball-reference.com/>

²See the website on <https://www.acb.com/>

Notice that the data encountered on the web is the average points per game and for one team, but we are interested in the total amount of points scored in a game. It is, we are not interested in the contribution of one team, but the contribution of both teams playing the game. Since in Table 1 we have the information per one team, we have decided to multiply all these numbers by two. Thus, the data that will be used is the numbers in the previous table multiplied by two. I have computed some summary statistics on these data. We have that the mean in the case of the NBA is 206.8 points per game, the minimum is 186.8 (in the 2003-04 season) and the maximum is 229.4 (the last season). In the case of the ACB the mean is 156.8, the minimum is 153 (in the 2013-14 season) and the maximum is 167 points (in the current season). Notice that our prior information is about the mean of the points per game. Thus, due the fact that $E(Y_1) = \lambda_{NBA}$ and $E(Y_2) = \lambda_{ACB}$, we have the prior information about the parameters $\lambda_{NBA}, \lambda_{ACB}$ directly. And, because of the minimums and maximums exposed before, we have decided to use the following prior information:

- λ_{NBA} is between 186.8 and 229.4 with a high probability, say 95%.
- λ_{ACB} is between 153 and 167 with a high probability, say 95%.

Using the well-known properties of the Gamma distribution we can choose the parameters of the prior distributions in a proper manner: if $Z \sim \text{Gamma}(\alpha, \beta)$ then

$$\begin{cases} E(Z) = \frac{\alpha}{\beta} \\ \sqrt{\text{Var}(Z)} = \frac{\sqrt{\alpha}}{\beta} \end{cases}$$

We can set up a system of equations in order to λ_{NBA} be centered on $(186.8 + 229.4)/2 = 208.1$ (the mid point of the interval (186.6, 229.4)) and with a standard deviation of $(229.4 - 208.1)/2 = 10.65$ (which let, approximately, the 95% of the probability within (186.6, 229.4)):

$$\begin{cases} \frac{\alpha_{NBA}}{\beta_{NBA}} = \frac{2081}{10} \\ \frac{\sqrt{\alpha_{NBA}}}{\beta_{NBA}} = \frac{213}{20} \end{cases}$$

The solution of the system above is $(\alpha_{NBA}, \beta_{NBA}) = (17322244/45369, 83240/45369)$. We do exactly the same in the ACB case, obtaining the system of equations:

$$\begin{cases} \frac{\alpha_{ACB}}{\beta_{ACB}} = 160 \\ \frac{\sqrt{\alpha_{ACB}}}{\beta_{ACB}} = \frac{7}{2} \end{cases}$$

Now the solution is $(\alpha_{ACB}, \beta_{ACB}) = (102400/49, 640/49)$. And then, we have the prior distributions and thus the Bayesian models completed.

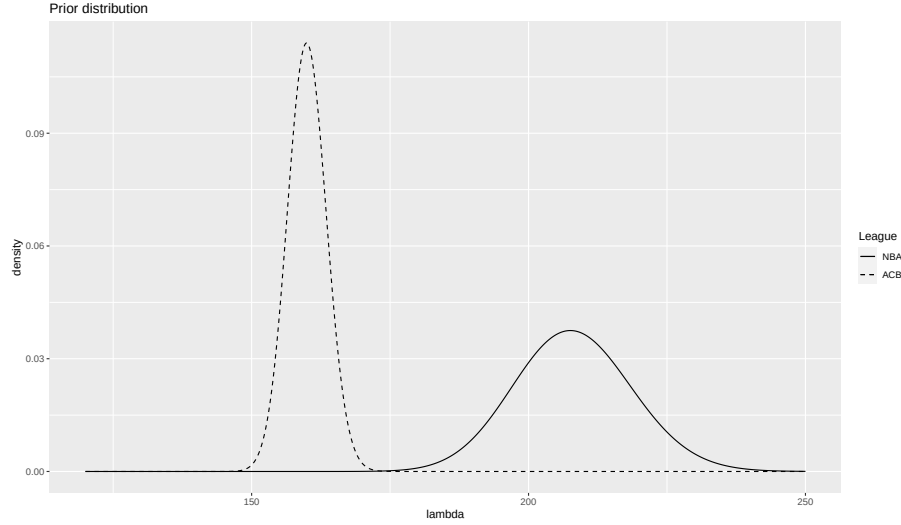


Figure 1: The prior distribution for both leagues.

- NBA: $Y_1 \sim Poiss(\lambda_{NBA})$, $\pi(\lambda_{NBA}) \sim Gamma(17322244/45369, 83240/45369)$.
- ACB: $Y_2 \sim Poiss(\lambda_{ACB})$, $\pi(\lambda_{ACB}) \sim Gamma(102400/49, 640/49)$.

Let us plot the prior distributions before go to the next section. The graphic with the prior distributions is exposes in the Figure 1. In the prior distributions graphics we observed what we were expecting, the scoring in the NBA is higher than in the ACB. In the next section we will see what the data has to say.

SECTION b): In the data file *Basquet.txt* you will find the total number of points of 20 matches taken randomly from ACB and 20 taken from NBA. Plot on the same graph the prior, the likelihood and the posterior distribution for each type of match.

SOLUTION:

We first present a table summarizing what we have until now.

League	Likelihood	Prior	Posterior
NBA	$Y_1 \sim Poiss(\lambda_{NBA})$	$\pi(\lambda_{NBA}) \sim G(\alpha_{NBA}, \beta_{NBA})$	$\lambda_{NBA} y_1 \sim G(\alpha_{NBA} + \sum y_1, \beta_{NBA} + n_1)$
ACB	$Y_1 \sim Poiss(\lambda_{ACB})$	$\pi(\lambda_{ACB}) \sim G(\alpha_{ACB}, \beta_{ACB})$	$\lambda_{ACB} y_2 \sim G(\alpha_{ACB} + \sum y_2, \beta_{ACB} + n_2)$

Table 2: Summary of the Bayesian models built.

In the Table 2 we have the likelihood, prior and posterior for each league. We have used that the Gamma distribution is the conjugate one in this case, and then we know the posterior distribution explicitly in advance. Nonetheless, when coding to perform the desired graphics,

we will be using the grid approach instead of the theoretical approach. As the reader know, there are several ways to do this. The reader will find the code used attached at the end of this report.

Having used the grid approach we have obtained the graphic presented in Figure 2. In this graphic we can observe that the posterior is, in both cases, a compromise between the prior and the likelihood. In addition, in the NBA case we can see that the likelihood is much more informative than our chosen prior. All the quantities exposed in the Figure 2 supports our previous idea that the scoring in the NBA is higher.

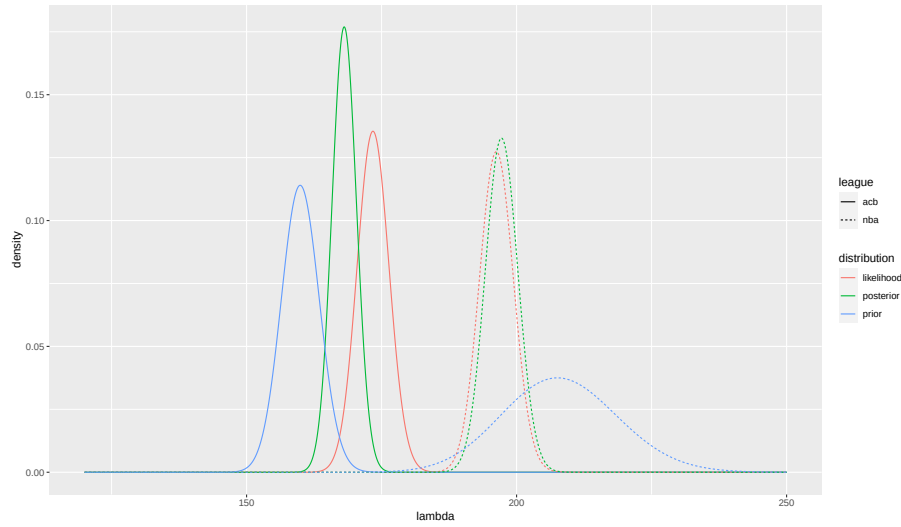


Figure 2: Prior, likelihood and posterior distribution for each league.

SECTION c): What is the probability that more points will be scored in a randomly selected match of the Spanish league than in a randomly selected match of the NBA league?

SOLUTION: In this section we should build a Bayesian hypothesis testing. We have to take into account that we are asked for points that *will be scored*. Hence, we have to work with the posterior predictive distribution which is

$$p(\tilde{y}_1|y_1) = \int p(\tilde{y}_1|\lambda_{NBA})\pi(\lambda_{NBA}|y_1)d\lambda_{NBA},$$

in the case of the NBA, and the analogous quantity in the case of the ACB. Note that all the components in the previous expression are well known, and then the posterior predictive distribution can be easily computed. In practice we will be using the simulation approach to do the computations (see the code in the end of the report).

The hypothesis testing that we will be using is:

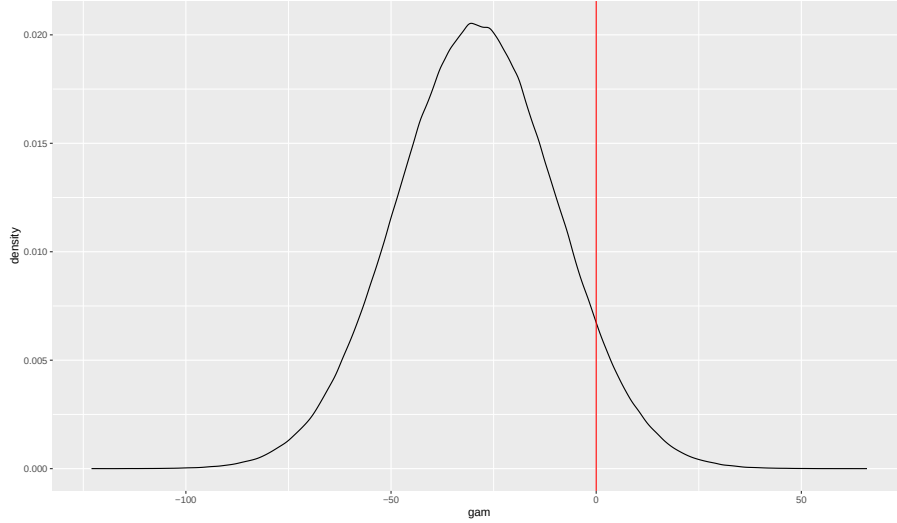


Figure 3: The γ computed as the difference of the posterior predictive distributions.

$$\begin{cases} H_1 : \tilde{y}_2 > \tilde{y}_1 \\ H_2 : \tilde{y}_2 \leq \tilde{y}_1 \end{cases} \iff \begin{cases} H_1 : \gamma = \tilde{y}_2 - \tilde{y}_1 > 0 \\ H_2 : \gamma = \tilde{y}_2 - \tilde{y}_1 \leq 0. \end{cases}$$

So, we can compute the probability $P(H_1|y_1, y_2) = P(\gamma > 0|y_1, y_2)$. Using the simulation approach, with a total amount of $M = 1000000$ simulations, we have obtained that the probability that more points will be scored in a randomly selected match of the ACB than in a randomly selected match of the NBA is 0.06349. It is,

$$P(H_1|y_1, y_2) = P(\gamma > 0|y_1, y_2) = 0.06349.$$

In addition, we have developed a graphic with the result shown in Figure 3. In this graphic we can see that the vertical line in the 0, which is colored in red. Looking at this graphic and this vertical line, we can have an insight on how γ behaves. In the code provided the reader can also see this problem solved using the theoretical approach, it is, using that we are working with the conjugate prior. And, indeed, the results obtained are very similar.

We have also computed the mean of the γ distribution, which is -29.15778 , meaning that in a one random match selected for the ACB and the NBA, on average, the score is -29.16 lower in the ACB than in the NBA.

SECTION d): What is the probability that the points scored in a NBA match will be 60 points or more larger than the points scored in an ACB match?

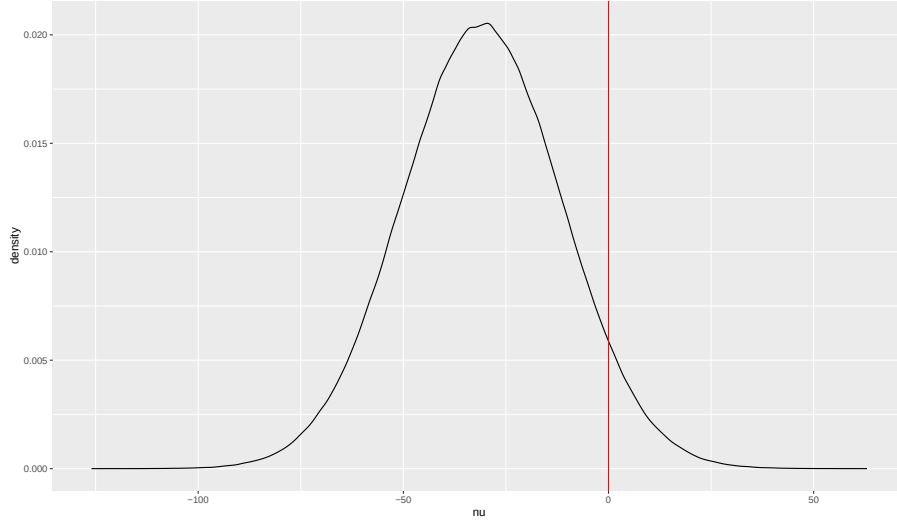


Figure 4: The ν distribution and the vertical line on the 0, in red.

SOLUTION: We will be following similar steps than in the previous section to develop this last part of the assignment. Basically, we are interested again in the posterior predictive distribution, which we have computed before. And, again, we want to build the proper hypothesis testing. In this case the hypothesis testing is:

$$\left\{ \begin{array}{l} H_1 : \tilde{y}_1 - \tilde{y}_2 > 60 \\ H_2 : \tilde{y}_1 - \tilde{y}_2 \leq 60 \end{array} \right\} \Longleftrightarrow \left\{ \begin{array}{l} H_1 : \nu = \tilde{y}_1 - \tilde{y}_2 - 60 > 0 \\ H_2 : \nu = \tilde{y}_1 - \tilde{y}_2 - 60 \leq 0. \end{array} \right.$$

So, in this case we have to compute the quantity ν , and then the interest relies on the probability $P(H_1|y_1, y_2) = P(\nu > 0|y_1, y_2)$. The computation of this quantity is developed as in the previous section, and we obtain:

$$P(H_1|y_1, y_2) = P(\nu > 0|y_1, y_2) = 0.053965.$$

The plot in this case is the one in Figure [4](#).

Homework II - Basketball

Arnau Garcia

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Homework II - Basketball

Section a):

We collect the average of points per game in the last twenty years of the NBA and the ACB, and we multiply by 2 because we are interested in the total amount of points scored per game.

```
nba<- c(114.5, 114.7, 110.6, 112.1, 111.8, 111.2, 106.3, 105.6, 102.7,
        100.0, 101.0, 98.1, 96.3, 99.6, 100.4, 100, 99.9, 98.7, 97.0, 97.2, 93.4)
acb <- c(83.5, 81.2, 78.4, 77.9, 76.5, 78.1, 81.8, 79.2, 77.8, 76.9,
        76.5, 77.4, 78.2, 77.9, 78.7, 77.8, 77.4, 78.1, 77.9, 77.3, 78.3)
nba <- 2*nba
acb <- 2*acb
```

We see now summary statistics of this data:

```
summary(nba)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   186.8   197.4   200.8   206.8   221.2   229.4
```

The mean of points in the last twenty years in the NBA is 206.8 points. The minimum is 186.8 (in the 2003-04 season) and the maximum is 229.4, which is the last season.

```
summary(acb)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   153.0   154.8   155.8   156.8   156.8   167.0
```

The mean of points in the last twenty years in the ACB is 156.8. The minimum is 153.0 (in the 2013-14 season) and the maximum is 167.0 points (in the current season).

The chosen parameters after solving the system of equations in the NBA case:

```
a_1 <- 17322244/45369
b_1 <- 83240/45369
```

Let us take a look in the probability between the desired values:

```
pgamma(229.4, shape = a_1, rate = b_1)-pgamma(186.8, shape = a_1, rate = b_1)
```

```
## [1] 0.9546404
```

Chosen parameters for the ACB case:

```
a_2 <- 102400/49
b_2 <- 640/49
```

The probability between the desired values:

```
pgamma(167, shape = a_2, rate = b_2) - pgamma(153, shape = a_2, rate = b_2)
```

```
## [1] 0.9545255
```

Section b)

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
library(purrr)
```

Let us read the data:

```
basket <- read.table("basquet.txt", header = TRUE)
```

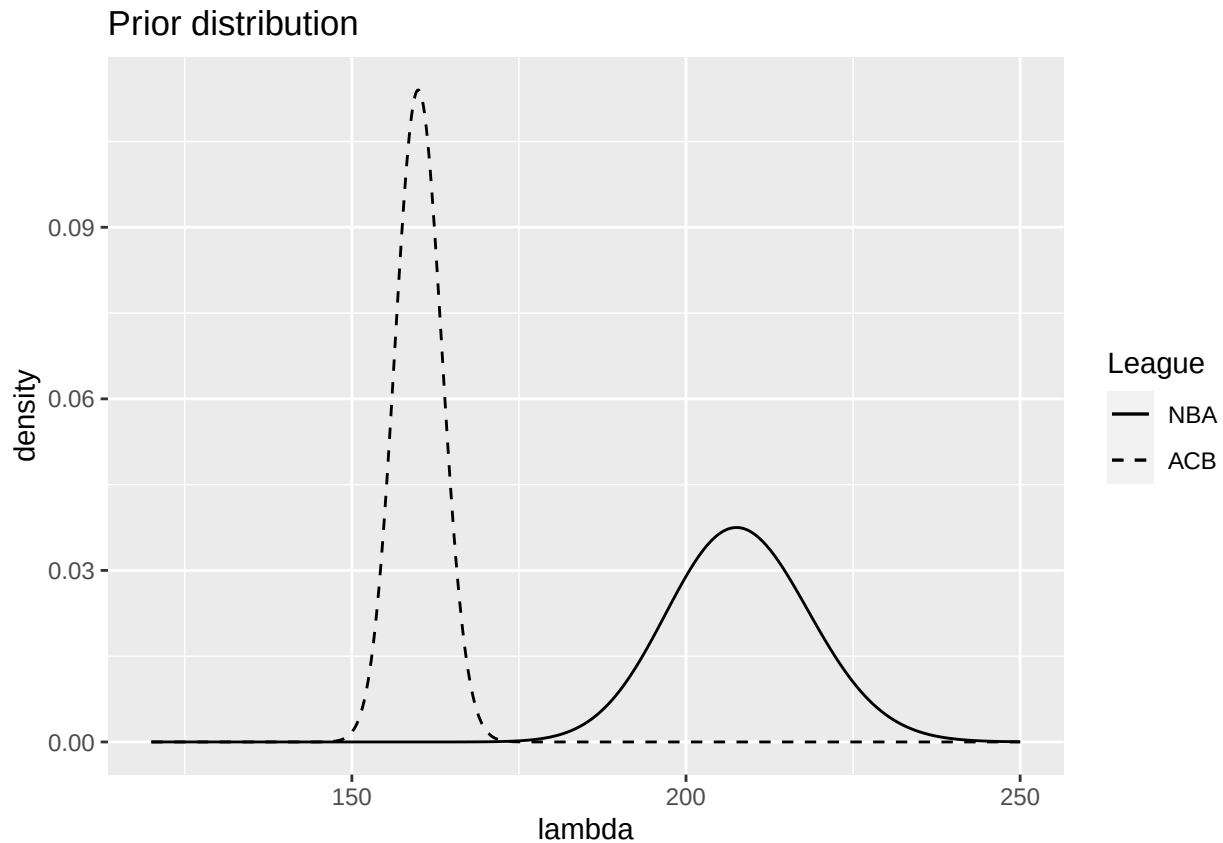
The priors:

```
## Prior distributions
delta_p <- 0.01
lambda <- seq(120, 250, delta_p)

# NBA prior:
prior_par_nba <- c(shape = a_1, rate = b_1)
prior_nba <- dgamma(lambda, shape=prior_par_nba[1], rate=prior_par_nba[2])

# ACB prior:
prior_par_acb <- c(shape=a_2, rate = b_2)
prior_acb <- dgamma(lambda, shape = prior_par_acb[1], rate = prior_par_acb[2])

# Plot prior comparison
df <- tibble(lambda, prior_nba, prior_acb)
ggplot(df) +
  geom_line(aes(x = lambda, y = prior_nba, linetype = "c")) +
  geom_line(aes(x = lambda, y = prior_acb, linetype = "e")) +
  ggtitle("Prior distribution") +
  ylab("density") +
  scale_linetype_manual(name = "League",
    values = 1:2, labels = c("NBA", "ACB"))
```



Now the priors likelihoods and posteriors together:

```
## Likelihood distributions
likelihood_nba <- map_dbl(lambda, ~ prod(dpois(basket$nba, .x)))
likelihood_acb <- map_dbl(lambda, ~ prod(dpois(basket$acb, .x)))

## Function to standardize
std_dist <- function(x, delta) {
  x / (sum(x) * delta)
}

## Posterior distributions
# Grid solution:
# Standardize likelihoods
# Calculate products
# Posteriors: standardize products
df <- df %>%
  mutate(likelihood_nba = std_dist(likelihood_nba, delta_p),
         likelihood_acb = std_dist(likelihood_acb, delta_p),
         product_nba = prior_nba * likelihood_nba,
         product_acb = prior_acb * likelihood_acb,
         posterior_nba = std_dist(product_nba, delta_p),
         posterior_acb = std_dist(product_acb, delta_p))

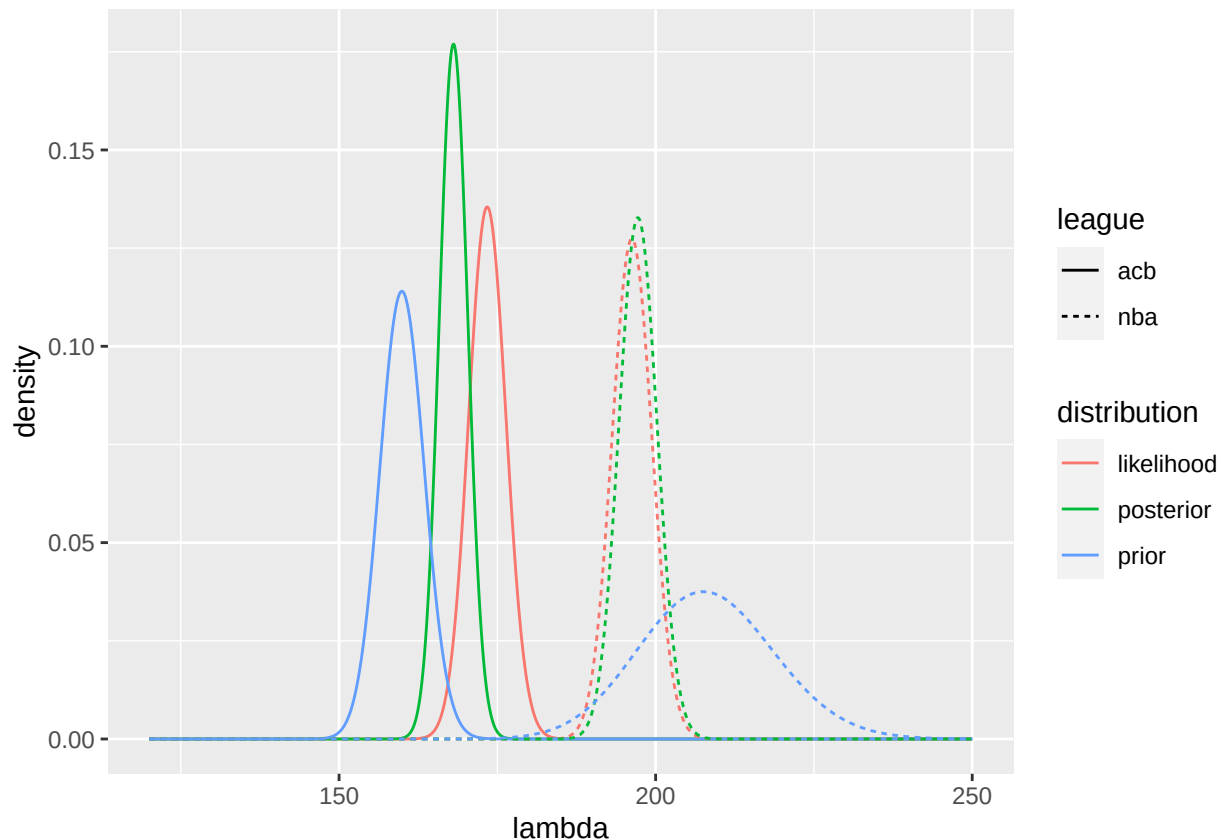
#Plot:
df_long <- df %>%
  select(-c(product_nba, product_acb)) %>%
  gather(distribution, density, -lambda) %>%
```

```

separate(distribution, into = c("distribution", "league"), sep = "_")

ggplot(df_long) +
  geom_line(aes(x = lambda, y = density, linetype = league, col = distribution))

```



In the code above we have used the grid solution, one can use the theoretical solutions using that we have worked with the conjugate distribution, and then the posterior is well-known.

Section c)

We use the simulation approach to simulate the posterior predictive distributions for each league:

```

M <- 1000000
post_sim_nba <- sample(x=lambda, size = M, replace = TRUE, prob = df$posterior_nba)
post_sim_acb <- sample(x=lambda, size = M, replace = TRUE, prob = df$posterior_acb)
pred_post_sim_nba <- rpois(M, post_sim_nba)
pred_post_sim_acb <- rpois(M, post_sim_acb)
gam <- pred_post_sim_acb - pred_post_sim_nba
mean(gam>0)

```

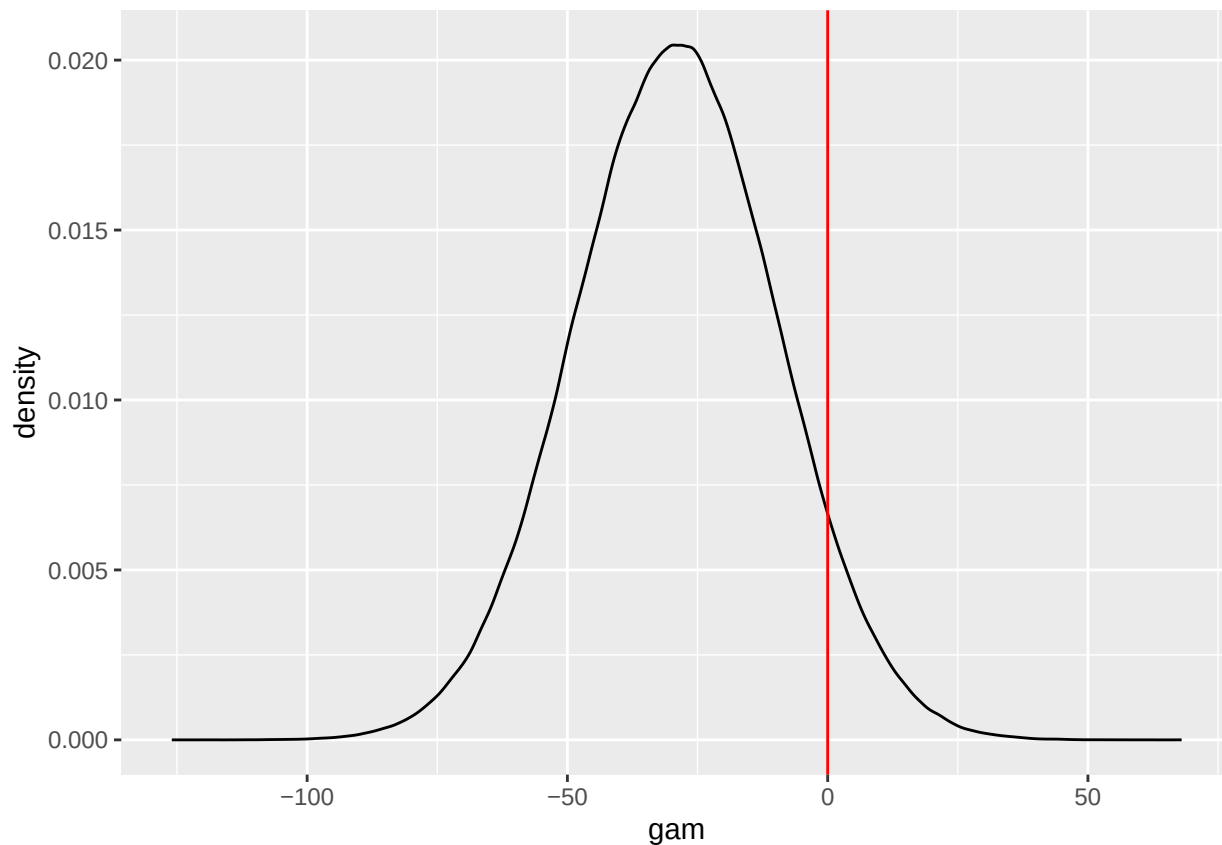
```
## [1] 0.063873
```

Now let us plot it

```

ggplot()+
  geom_density(data = tibble(gam), aes(gam))+
  geom_vline(xintercept = 0, color="red")

```



In addition we can compute:

```
mean(gam)
```

```
## [1] -29.15049
```

Meaning that in a one random match selected for the ACB and the NBA, on average, the score is -29.16 lower in the ACB than in the NBA.

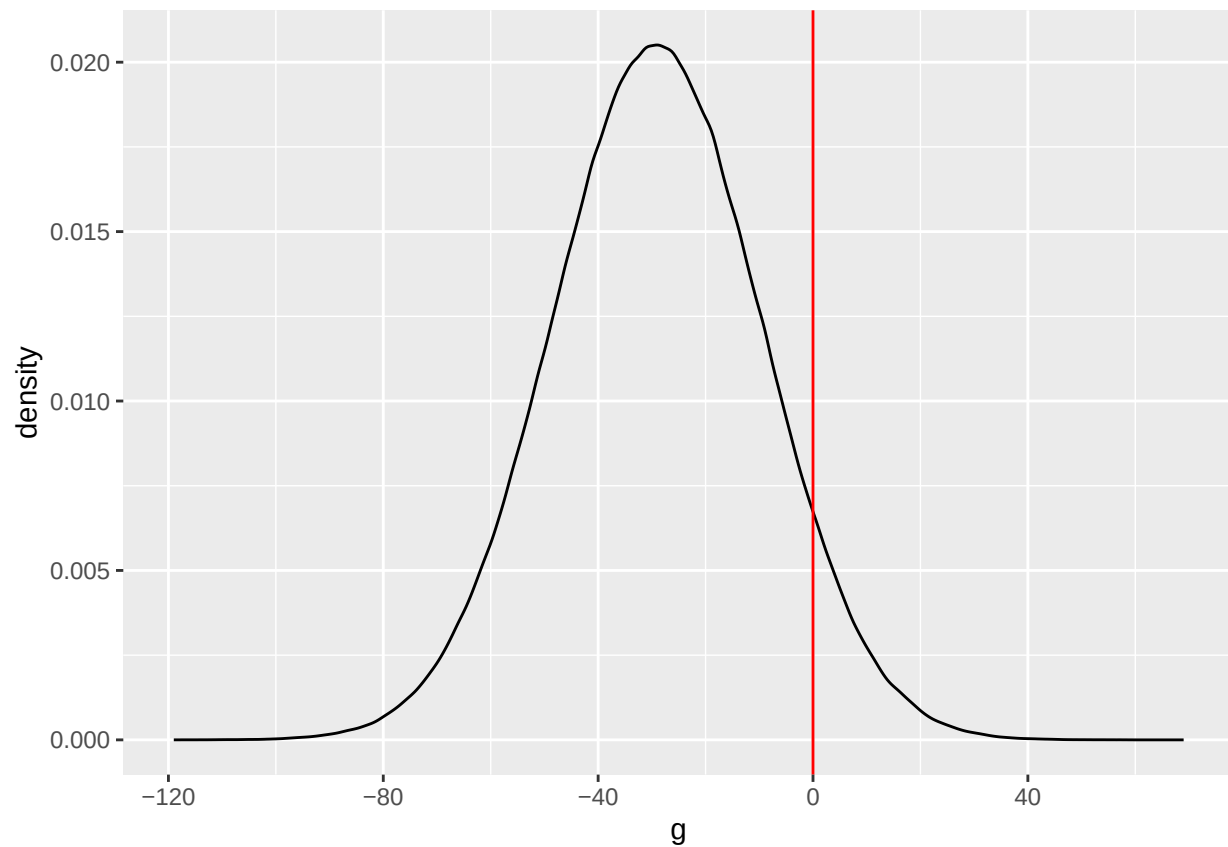
Theoretical option (see the properties of our conjugate prior on (https://en.wikipedia.org/wiki/Conjugate_prior)):

```
pred_post_sim_nba_theo <- rnbinoM(M, size = a_1+sum(basket$nba),
                                prob = (b_1+length(basket$nba))/(1+b_1+length(basket$nba)))
pred_post_sim_acb_theo <- rnbinoM(M, size = a_2+sum(basket$acb),
                                prob = (b_2+length(basket$acb))/(1+b_2+length(basket$acb)))
g <- pred_post_sim_acb_theo-pred_post_sim_nba_theo
mean(g>0)
```

```
## [1] 0.064275
```

Now the plot:

```
ggplot()+
  geom_density(data = tibble(g), aes(g))+
  geom_vline(xintercept = 0, color="red")
```



Indeed the results obtained are very similar.

Section d)

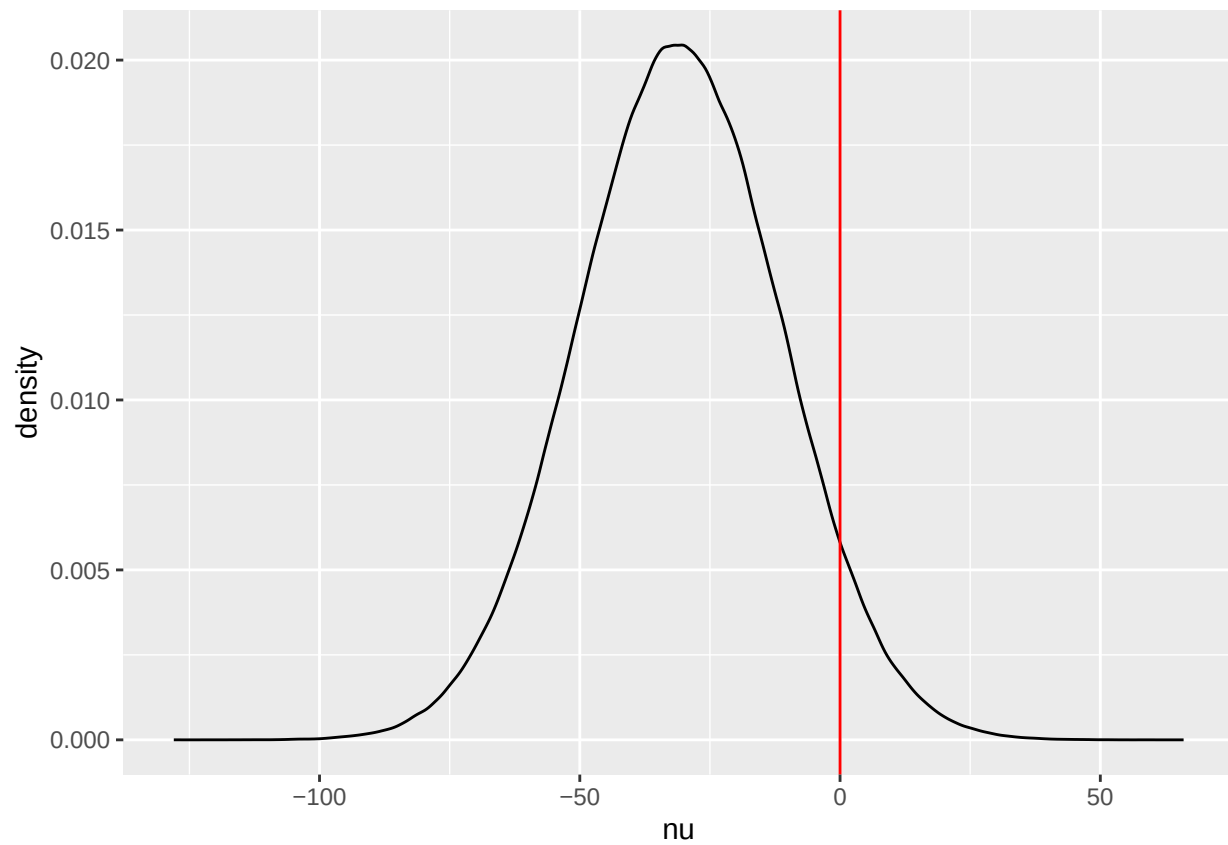
Let us compute the ν quantity:

```
nu <- pred_post_sim_nba - pred_post_sim_acb - 60
mean(nu>0)
```

```
## [1] 0.053935
```

And finally the plot:

```
ggplot()+
  geom_density(data = tibble(nu), aes(nu))+
  geom_vline(xintercept = 0, color="red")
```



Theoretical alternative:

```
mean(pred_post_sim_nba_theo-pred_post_sim_acb_theo-60>0)
```

```
## [1] 0.053789
```

Hence, we have a similar result as expected.